

Pranav Verma

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Senior Perception Engineer | Real-Time Localization, Mapping, 3D Computer Vision, and Production ML Systems

WORK EXPERIENCE

Senior Software Engineer, Perception, Magic Leap

June 2021 - Present

- **Designed and shipped the C++ On-Device Localization pipeline**, enabling **real-time spatial persistence** and **visual localization** for Magic Leap 2
- **Integrated, benchmarked, and deployed learned feature extraction and image retrieval models**, reducing localization **pose error** by **15–26%**.
- **Built automated eval frameworks** for production localization pipelines against GT poses via **nightly device tests** and **hardware-in-the-loop validation**.
- **Designed and integrated persistence and VPS perception** algorithms for **Android and iOS-based AR glasses** in collaboration with partner companies.
- Redesigned the localization pipeline to **reduce memory usage by 12x**, freeing **25% of on-device memory** for other runtime features.
- Re-architected a high-frequency on-device perception workload into an event-driven pipeline, **reducing compute cost by 94%**.
- **Reduced localization latency by 60%+** through loading optimizations, caching, and runtime profiling, while preserving accuracy targets.
- Developed localization consistency algorithms for large-scale spaces, **reducing visible pose instability by 70%**.
- **Led technical design reviews** and **cross-functional debugging** of production issues across perception, platform, QA, product, and customer teams, using logs, Perfetto traces, benchmark regressions, and hardware-in-the-loop reproductions.

Graduate Student Researcher, Prof. Falko Kuester, UC San Diego

October 2019 - June 2021

- Built **Dockerized** AirSim, Unreal Engine, and CARLA 3D simulation pipelines for **scalable synthetic data generation**, supporting **reproducible ML training and evaluation** for UAV reinforcement learning, motion segmentation, scene flow, and point-cloud tasks.
- Developed and evaluated unsupervised/semi-supervised CV models, labeled-data pipelines, and domain-adaptation workflows for real-time smoke localization across 600+ deployed cameras.
- Designed a **multi-view camera calibration** method achieving **sub-pixel accuracy** for multi-camera computer vision systems; published in 3D Imaging and Applications, 2021. [\[Paper\]](#)
- Developed a streaming-based demosaicing and rectification algorithm for FPGA deployment, achieving comparable reconstruction quality with **10x lower memory usage**; published at WACV 2021. [\[Project page + Code\]](#)

Software Engineering Intern, Texas Instruments, Bangalore

May - July 2018

- Developed a robust, unit-tested computer vision tool using OpenCV to automate calibration of TI's 4K projector chips, **reducing latency by 80%**.
- Engineered calibration algorithms robust to lighting variations and low-resolution cameras; the tool was adopted on the production factory floor.

TECHNICAL SKILLS

Languages: C++, Python, C, Bash, MATLAB | **Libraries / Tools:** PyTorch, TFLite, OpenCV, Eigen, Ceres, COLMAP, Abseil, Git, g2O, CMake, Bazel, Docker

Robotics / Localization: visual localization, pose estimation, relocalization, spatial transforms, spatial persistence, hardware-in-the-loop validation

Perception / Geometry: multi-view geometry, camera calibration, feature extraction and matching, image retrieval, nonlinear least squares, sensor / camera modeling, real-time perception

Simulation / Evaluation: CARLA, AirSim, Unreal Engine, synthetic data generation, benchmark regression analysis, ground-truth evaluation, real-time perception evaluation

Systems / Optimization: embedded systems, Linux, Android, iOS, runtime tracing, latency / memory / compute optimization, Perfetto, pprof

EDUCATION

University of California, San Diego

2019-2021

Master of Science (M.S.), Computer Science and Engineering (Focus: Graphics and Computer Vision)

GPA: 3.88/4.0

Indian Institute of Technology, Delhi

2015-2019

Bachelor of Technology (B.Tech.), Electrical Engineering (Power & Automation)

GPA: 8.541/10.0

PUBLICATIONS

- [1] Dominique E. Meyer, **Pranav Verma**, and Falko Kuester. "3D Multi-Camera Calibration with a Fractal Encoded Multi-Shape Target". In: *Electronic Imaging* 33.18 (2021), pp. 101-1–101-1. [\[Paper\]](#).
- [2] **Verma, Pranav** et al. "Splatty: A Unified Image Demosaicing and Rectification Method". In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 2021, pp. 786–795. [\[Paper\]](#).

PROJECTS

Pratyaksh: Vibro-Tactile Feedback for the Blind Prof. M. Balakrishnan, CSE, IIT Delhi

January 2017- June 2017

- Built an embedded armband wearable and companion Android app, implemented C++ computer vision algorithms for real-time processing of depth maps from Intel RealSense sensors for spatial perception and obstacle avoidance. [\[Link\]](#)